HYPERTONIC SALINE 3%

High-alert drug

Correct Na^+ ≤8–10 mmol/L/24 h

CHEMICAL NAME NaCl 30 g/L in water — Na^+ 513 mmol/L; osmolarity ~1,026 mOsm/L

CLASS Hypertonic crystalloid (concentrated electrolyte)

TRADE NAMES Sodium chloride 3% · Hypertonic saline · (Also 7% nebulised for cystic fibrosis)

THERAPEUTIC USES / INDICATIONS

- Severe symptomatic hyponatraemia (seizures, coma, severe confusion, vomiting).
- Raised intracranial pressure (alternative to mannitol), cerebral oedema.
- Nebulised (3–7%) in bronchiolitis (some units) and cystic fibrosis.

ADMINISTRATION

- **IV** via large peripheral vein or central line using infusion pump; clearly labelled.
- Store separately from 0.9% saline.
- Frequent sodium checks (every 2–4 h) in hyponatraemia.
- ICU/HDU setting usually.

DOSAGES

- **Severe symptomatic hyponatraemia:** IV bolus of 3% saline at once (senior decision, local protocol), repeated with Na^+ checks; aim for a 5 mmol/L rise in the first hour.
- **Raised ICP:** used as osmotherapy per neurocritical care protocol (specialist).
- **Correction limit:** no more than 10 mmol/L in the first 24 h and 18 mmol/L in 48 h; stop hypertonic fluid if exceeded.

PHARMACOKINETICS

- Remains largely extracellular; draws water from cells (osmotic effect).
- Excreted by kidneys.

PHARMACODYNAMICS

- Raises **serum sodium and osmolality** → water shifts out of brain cells → ↓ cerebral oedema and ICP.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis, pain at infusion site.
- Thirst.

ADVERSE EFFECTS (SERIOUS)

- Osmotic demyelination syndrome (central pontine myelinolysis); dysarthria, dysphagia, quadriparesis, locked-in — from overly rapid correction.
- Fluid overload, pulmonary oedema.
- Hypernatraemia, hyperchloraemic acidosis; rebound ICP.

CONTRAINDICATIONS & CAUTIONS

- Hypernatraemia; fluid overload.
- Asymptomatic or chronic mild hyponatraemia (treat cause).
- **Caution:** heart failure.

KEY INTERACTIONS

- Desmopressin, diuretics affect sodium correction rate; coordinate therapy.

NURSING MANAGEMENT

- Serum Na^+ every 2–4 h in hyponatraemia; strict fluid balance, urine output (overcorrection with brisk diuresis).
- Neurological observations (GCS, pupils).
- Check site frequently.
- Escalate if Na^+ rises too fast (may need desmopressin + D5W).

PATIENT EDUCATION

- Explain frequent blood tests are needed to correct sodium safely.

KEY POINTS

- Hypertonic saline — 513 mmol/L Na^+.
- Max correction 10 mmol/L in 24 h (osmotic demyelination).
- Severe symptoms: bolus, aim Na^+ +5 mmol/L in 1 h.
- Also for raised ICP.

SOURCES CHECKED **[956]** Sodium Chloride 30%w/v Concentrate for Solution for Infusion (UK SmPC) **[957]** Ball et al., Endocr Connect 2016 *Full references p. 432–457*